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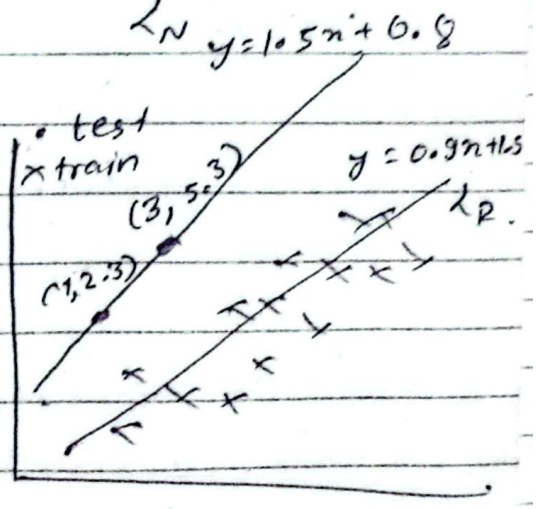
we have to do little bit low value of m for regularization.

Here we have to choose L_R.



$$\text{Total Loss} = \left\{ \underbrace{\frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n}}_{\text{loss}} + \lambda \cdot \|m^2\| \right\}$$

Here, λ = hyper parameter $[\lambda \in (0, \infty)]$.

step 1: Calculate loss for both lines.

Let $\lambda = 1$ [for now].

Loss L_N	Loss L_R
$(0 + 1.5^2)/6$	$(2.9 - 0.9 - 1.5)^2 +$
$= 2.25$	$(5.2 - 2.7 - 1.5)^2$
	$= 0.1^2 + 1.1^2 + 0.9^2$
	$= 2.03$.

so, ml will choose L_R coz of regularization.

~~Use~~ Prefer regularization to linear regress... when high multicollinearity, enough data ($n >> p$) (1000, 5)

- Use when ~~data~~ Dimension > 2

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* Here we multiply with L2 norm

for $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$

$$\lambda . \| \beta_1^2 + \beta_2^2 + \beta_3^2 \|$$

```
from sklearn.linear_model import Ridge.
R = Ridge( alpha = 0.001)
```

↓ check & do hyperparameter tuning.

```
R.fit (X_train, ~~X_test~~ y_train)
y_pred1 = R.predit (X_test).
```

R.coef_ R.intercept_

Math

$$\text{Total Loss} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda . m^2$$

$$= \sum_{i=1}^{n} (y_i - mx_i - b)^2 + \lambda . m^2$$

we got : $b = \bar{Y} - m\bar{X}$ } same as befor.

For m :

$$L = \sum_{i=1}^{n} (y_i - mx_i - \bar{y} + m\bar{x}) + \lambda . m^2 .$$

For

$$\frac{\partial L}{\partial m} = 0$$

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$$\frac{\partial L}{\partial m} = 2 \cdot \sum_{i=1}^{n} (y_i - m x_i - \bar{y} + m\bar{x})(-x_i + \bar{x}) + 2\lambda m$$

$$\Rightarrow 0 = -2 \sum_{i=1}^{n} (y_i - m x_i - \bar{y} + m\bar{x})(x_i - \bar{x}) + 2\lambda m$$

$$\Rightarrow \lambda m - \sum_{i=1}^{n} \left[(y_i - \bar{y}) - m(x_i - \bar{x}) \right] (x_i - \bar{x}) = 0$$

$$\Rightarrow \lambda m - \sum_{i=1}^{n} \left[(y_i - \bar{y})(x_i - \bar{x}) - m(x_i - \bar{x})^2 \right] = 0$$

$$\Rightarrow \lambda m + m \sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x})$$

$$\Rightarrow \boxed{m = \frac{\sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^{n} (x_i - \bar{x})^2 + \lambda}}$$ Ridge Regression.

$$\boxed{m = \frac{\sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}}$$ ← linear regression.

Here λ = alpha.

For, $\lambda = 0 \rightarrow$ Ridge = Linear.

High λ, $m \downarrow$

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For multi dimension.

$$L = \sum_{i=1} (y_i - \hat{y}_i)^2$$

$$= (XW - Y)^T (XW - Y)$$

$$Y = \begin{bmatrix} - \\ - \\ - \\ - \end{bmatrix} \quad W = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix}_{(n+1)\times 1} \quad X = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1n} \\ 1 & x_{21} & x_{21} & \cdots & x_{2n} \\ \vdots & \vdots & & & \\ 1 & x_{m1} & \cdots & & x_{mn} \end{bmatrix}$$

$$L = (XW - Y)^T (XW - Y) + \lambda . \|W\|^2$$ shrinkage coefficient.

$$\Downarrow$$

$$\{\lambda w_0^2 + \lambda w_1^2 + \lambda w_2^2 + \lambda w_3^2 - \cdots\}$$

$$\lambda . W^T . W$$

$$\boxed{L = (XW - Y)^T (XW - Y) + \lambda . W^T W}$$

$$L = [(XW)^T - Y^T][XW - Y] + \lambda . W^T W.$$

$$\left\{ \begin{array}{l} (a-b)^T = a^T - b^T \\ (ab)^T = b^T a^T \end{array} \right\}$$

$$= (W^T X^T - Y^T)(XW - Y) + \lambda . W^T W$$

$$= W^T X^T X W - W^T X^T Y - Y^T X W + Y^T Y + \lambda W^T W$$

$$L = W^T X^T X W - 2W^T X^T Y + Y^T Y + \lambda W^T W$$

CamScanner

(use formula in internet & do)

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$$\frac{\partial L}{\partial w} = 2X^TXw - 2X^TY + 0 + 2\lambda w = 0$$

$$\Rightarrow X^TXw + \lambda w = X^TY$$

$$\Rightarrow \boxed{w = (X^TX + \lambda I)^{-1} X^TY} \text{ ridge}$$

$$w = (X^TX)^{-1} X^TY \text{ } \} \text{ linear}$$

To insert 1 in each row:

```
np.insert(X_train, 0, 1, axis=1)
```

```
I = np.identity(X_train.shape[1])
```

shape (1000, 5)

(5x5)

```
w → ~~result~~ result = np.linalg.inv(np.dot(X_train.T,
        X_train) + self.alpha * I).dot(X_train.T).
        dot(y_train)

self.inter = result[0]
self.coef = result[1:].
```